

## Scan Your Local Network for Open Ports

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Tast # 1

**Objective: Learn to discover open ports on devices in your local network to understand network exposure**

- **Install Nmap from official website.**

Installing nmap using linux cmd command.

```
└─$ sudo apt install nmap
```

[sudo] password for kali:

nmap is already the newest version (7.99+dfsg-1kali1).

The following packages were automatically installed and are no longer required:

|                                      |                       |                     |                       |
|--------------------------------------|-----------------------|---------------------|-----------------------|
| amass-common                         | libgdata22            | libportmidi0        | libzxing3             |
| python3-kismetcapturefreaklabszigbee |                       |                     |                       |
| curlftps                             | libgdk-pixbuf2.0-bin  | libpostproc58       | node-async-each       |
| python3-kismetcapturertl433          |                       |                     |                       |
| enchant-2                            | libgeos3.13.1         | libqt5ct-common1.8  | node-cjs-module-lexer |
| python3-kismetcapturertladsb         |                       |                     |                       |
| firmware-ti-connectivity             | libgirepository-1.0-1 | libradare2-5.0.0t64 | node-core-util-is     |
| python3-kismetcapturertlamr          |                       |                     |                       |
| gir1.2-girepository-2.0              | libgpgme11t64         | libraw23t64         | node-depd             |
| python3-packaging-whl                |                       |                     |                       |
| libabsl20240722                      | libgpgmepp6t64        | librubberband2      | node-fs.realpath      |
| python3-pluginbase                   |                       |                     |                       |
| libann0                              | libhdf4-0-alt         | libsframe1          | node-inflight         |
| python3-protobuf                     |                       |                     |                       |
| libarmadillo14                       | libinstpatch-1.0-2    | libsigsegv2         | node-ip               |
| python3-pysmi                        |                       |                     | python3-              |
| libaudio2                            | libjs-is-typedarray   | libsnmp40t64        | node-is-binary-path   |
| python3-pysnmp4                      |                       |                     |                       |
| libavc1394-0                         | libjs-jquery-ui       | libsodium23         | node-is-typedarray    |
| python3-terminaltables               |                       |                     |                       |

|                                         |                                                 |                   |                               |        |
|-----------------------------------------|-------------------------------------------------|-------------------|-------------------------------|--------|
| libavfilter10<br>python3-tz             | libjs-source-map                                | libsoup-2.4-1     | node-path-is-absolute         |        |
| libavformat61<br>args                   | libjs-typedarray-to-buffer<br>python3-wheel-whl | libsoup2.4-common | node-process-nextick-args     |        |
| libblosc2-4<br>python3-xlrd             | libjs-underscore                                | libsphinxbase3t64 | node-readdirp                 |        |
| libbluray2<br>python3-xlutils           | liblab-gamut1                                   | libsqlcipher1     | node-sellside-emitter         |        |
| libbson-1.0-0t64<br>python3-xlwt        | libmbedcrypto16                                 | libswscale8       | node-set-immediate-shim       |        |
| libcdt5<br>python3-yaswfp               | libmjpegutils-2.1-0t64                          | libteamctl0       | node-typedarray-to-buffer     |        |
| libcgraph6<br>python3-zombie-imp        | libmongoc-1.0-0t64                              | libtheora0        | node-uri-js                   |        |
| libconfig-inifiles-perl<br>ruby-unf-ext | libmpeg2encpp-2.1-0t64                          | libtsk19t64       | node-uuid                     |        |
| libcrypt-dev<br>samba-ad-dc             | libmplex2-2.1-0t64                              | libudfread0       | openjdk-21-jre                |        |
| libdisplay-info2<br>samba-ad-provision  | libmupdf25.1                                    | libunibreak6      | openjdk-21-jre-headless       |        |
| libdlt2<br>dsdb-modules                 | libnet1                                         | libvdpau-va-gl1   | pocketsphinx-en-us            | samba- |
| libfuse2t64<br>driver-all               | libogdi4.1                                      | libwireshark18    | python3-bluepy                | vdpa-  |
| libgav1-1                               | libplacebo349                                   | libwiretap15      | python3-click-plugins         |        |
| libgdal36                               | libpocketsphinx3                                | libwsutil16       | python3-fs                    |        |
| libgdata-common                         | libpoppler147                                   | libyelp0          | python3-kismetcapturebtgeiger |        |

Use 'sudo apt autoremove' to remove them.

Summary:

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0

- **Find your local IP range**

Using command ip a my linux machine ip is 192.168.100.189

ip a

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group
default qlen 1000
```

```
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

```
inet 127.0.0.1/8 scope host lo
```

```
valid_lft forever preferred_lft forever
```

```
inet6 ::1/128 scope host noprefixroute
```

```
valid_lft forever preferred_lft forever
```

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
```

```
link/ether 00:0c:29:9c:f4:be brd ff:ff:ff:ff:ff:ff
```

```
inet 192.168.100.189/24 brd 192.168.100.255 scope global dynamic noprefixroute eth0
```

```
valid_lft 83220sec preferred_lft 83220sec
```

```
inet6 fe80::dfdc:cb3c:cedc:171f/64 scope link noprefixroute
```

```
valid_lft forever preferred_lft forever
```

- **Run: nmap -sS 192.168.1.0/24 to perform TCP SYN scan.**

First discovered the open host, for **192.168.1.0/24**, then executed the command nmap -sS for open hosts,

```
(kaliⓈkali)-[~]
```

```
└─$ nmap -sS --max-retries 2 192.168.1.85 192.168.1.13 192.168.1.9 192.168.1.5
```

Starting Nmap 7.99 ( <https://nmap.org> ) at 2026-06-29 05:36 -0400

Warning: 192.168.1.13 giving up on port because retransmission cap hit (2).

Warning: 192.168.1.9 giving up on port because retransmission cap hit (2).

Warning: 192.168.1.5 giving up on port because retransmission cap hit (2).

Warning: 192.168.1.85 giving up on port because retransmission cap hit (2).

Testing on linux machine ip **192.168.100.189**

└─(kali㉿kali)-[~]

└─\$ nmap -sS 192.168.100.189

Starting Nmap 7.99 ( <https://nmap.org> ) at 2026-06-29 05:05 -0400

Nmap scan report for 192.168.100.189

Host is up.

All 1000 scanned ports on 192.168.100.189 are in ignored states.

Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 201.94 seconds

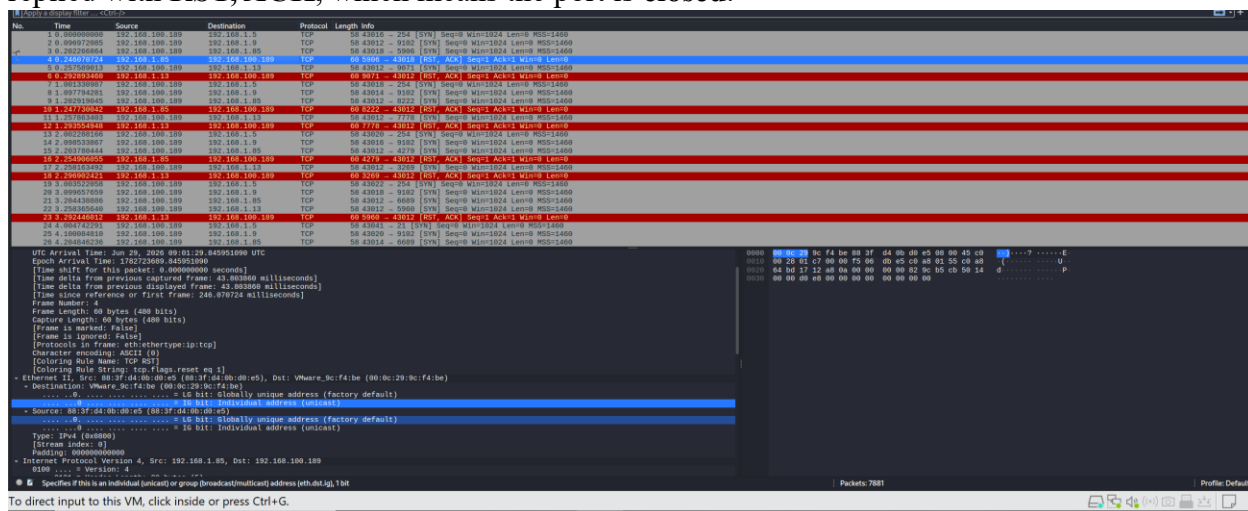
- **Note down IP addresses and open ports found.**

For IP 192.168.100.189 1000 ports were scanned but all ports were filtered. All scanned TCP ports are being filtered by a firewall or packet filter, so Nmap received no responses.

- **Optionally analyze packet capture with Wireshark.**

Analyzing the packet using wireshark the target port is filtered (or the response was not captured), preventing completion of the TCP handshake.

two packets together for **TCP SYN scan against port 5906**. The target immediately replied with **RST, ACK**, which means the port is **closed**.



- **Research common services running on those ports.**

Tested the IP but open ports were found all the ports were filtered or firewall dropping the packets. The output shows that Nmap is still not receiving responses from the target hosts, even after reducing the retry limit.

Nmap did not receive a response after the configured number of retransmissions, so it stopped probing that port. This typically indicates packet filtering by a firewall or another device that silently drops the probes.